

AMRUTH SHYJU

MERN Stack Developer — React.js — Node.js — MongoDB — TypeScript

amruthshyju@gmail.com | +91 8590797504 | Portfolio | LinkedIn | GitHub

Professional Summary

Backend Engineer specializing in building **high-throughput distributed systems** using **Node.js** and **TypeScript**. Experienced in architecting **real-time event-driven systems**, queue-based processing pipelines, and scalable multi-tenant backend platforms with sub-100ms communication latency. Strong focus on **Clean Architecture**, modular system design, and production-grade backend engineering.

Technical Skills

Languages: JavaScript (ES6+), TypeScript
Backend: Node.js, Express.js, REST APIs, Clean Architecture, Dependency Injection
Distributed Systems: Redis, BullMQ, Event-Driven Architecture, Asynchronous Processing
Real-Time Systems: Socket.IO, WebSockets, Pub/Sub Messaging
Databases: MongoDB (Transactions, Indexing, Data Modeling), SQL
Frontend: React.js, Next.js, TanStack Query, Leaflet
DevOps & Cloud: Docker, Nginx, AWS EC2, CI/CD
Security: JWT Authentication, RBAC, Rate Limiting, Zod Validation
Tools: Git, GitHub, Postman, Jira, Figma

Projects

SPEEDO – GPS Fleet Trip Analytics System Live | GitHub

Node.js, TypeScript, MongoDB, Redis, BullMQ, Socket.IO, React, Leaflet

- Architected a scalable real-time telemetry pipeline using **BullMQ, Redis, and Socket.IO** to process high-frequency GPS events with reliable asynchronous job handling, retry strategies, and low-latency live vehicle synchronization.
- Designed a clean-architecture backend in **TypeScript** with dependency injection, interface-driven service layers, and **MongoDB transactions**, enabling modular, testable, and atomic trip-processing workflows across 70+ interconnected modules.
- Developed geospatial analytics features including **trip tracking, overspeed detection, stoppage analysis, and driver behavior insights** using optimized **O(n) streaming algorithms** and haversine-based distance computations, reducing trip processing latency by **70%**.
- Built a production-grade **React 19 + Leaflet** GIS dashboard with real-time Socket.IO synchronization, efficient rendering of thousands of GPS points, and TanStack Query caching for responsive live fleet monitoring.

STRATIFY – Multi-Tenant ERP Platform Live | GitHub

Node.js, TypeScript, React, MongoDB, Redis, Socket.IO, AWS

- Architected a **scalable multi-tenant backend platform** using **Clean Architecture, TypeScript, and tsyringe-based dependency injection**, enforcing strict separation of concerns and highly modular service design.
- Designed a production-grade **RBAC authorization system** with **JWT authentication**, tenant-level access control, and centralized **Zod-based request validation**, ensuring secure and isolated multi-tenant workflows.
- Engineered **real-time collaboration infrastructure** using **Socket.IO + Redis Pub/Sub**, enabling low-latency multi-room messaging, live notifications, and scalable event-driven communication across distributed services.
- Integrated third-party services including **Razorpay, Cloudinary, and ZegoCloud** through a reusable **service abstraction layer**, improving extensibility, maintainability, and domain-layer independence.

Experience

Software Development Trainee (MERN Stack) Brototype

Industry Training Program

2024 – Present

- Developed scalable backend services using **Node.js** and **TypeScript**, handling both **real-time communication** and asynchronous event-driven workloads.
- Architected modular backend systems using **Clean Architecture**, enabling maintainable codebases with clear separation between domain, application, and infrastructure layers.
- Contributed to backend system design decisions involving **WebSockets, Redis, queue-based processing, and scalable communication patterns** to improve reliability and performance.

Education

Higher Secondary Education (Computer Science)

St. Catherine's HSS, Kerala State Board

2022